Methods and Systematics — AGEL0206 σ_e ApJL paper

Last updated: 2026-05-18 (narrative); headline refreshed 2026-06-16 (M★ audit).

> ⚠ CURRENT HEADLINE (2026-06-16, M★ audit: validated-fit apcorr + quiescent SED prior) — authoritative source **`results/PAPER_VALUES.json`**: σ_e(<R_e) = 267.31 ± 11.77 km/s (asym −11.98/+11.58) · log M★/M⊙ = 11.5 ± 0.1 (stat) ± 0.2 (sys) (2dp 11.46) · R_e = 2.097″ = 15.26 kpc · z_defl = 0.67564 · z_src = 1.30263. (M★ was 11.50 pre-audit; see DRAFTING §3.2.) The narrative BODY below predates the M8–M12 mask/systematic audits and prints earlier values (254.85 / 268.98, total ±18) in prose — those are HISTORICAL. For current numbers read this banner, `CLAUDE.md`, or `DRAFTING_FACTS_paper_2026-05-29.md`; all are emitted by `scripts/paper_values.py`.

This document is a single-pass narrative of the spectroscopic (σ_e) and photometric (R_e, M★) pipelines used for the AGEL J020613−011417 deflector, with the systematic-error budget worked into the text rather than relegated to a separate appendix. It is intended to:

- serve as a methods-section draft for the ApJL paper;
- give a new collaborator (or a future Claude session) a complete picture of how every headline number was produced and what was checked;
- complement `TESTS_AND_DIAGNOSTICS.md`, which is the long-form catalog of every individual test (A1–L3) with status and cache paths.

Where a test code (e.g. D3, J5) appears in brackets, it points to the matching row of `TESTS_AND_DIAGNOSTICS.md` §0 for the full provenance.

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0. Headline numbers

Quantity	Value	Source / notebook
σ_e(<R_e) — paper headline, wide arc-masked window, NEW `_mtwdo_` reduction, bad-pixel mask + Balmer kept + He I 3819 mask + M10 full sky audit + M11 rigorous re-derivation of carried systematics	267.31 ± 11.77 km/s (best-mask R_e=2.097″; symmetric; asymmetric form −11.98 / +11.58 preserves the stat-side bootstrap skew). M11 (2026-05-27) replaced the carried I-shape (±1.5), F200-mask (±3.8), and fit-window (±15) components with cube+mask-matched values (±2.27 / ±6.65 / ±3.82). The fit-window drop is the dominant change (the 3 windows now agree to ±4 km/s on the cleaned NEW cube). Reduction-pass component ±3.45 (M10).	`scripts/run_wide_sigma_e.py --cube new_clean_hei`, caches at `results/run_wide_sigma_e/new_clean_hei/`. M11 sweep caches at `results/{ishape_sweep,maskweight_sweep}_wR3800_5400_arcmask_M10/` + `results/nb09a_wavelength_sweep_M10/`
σ_e(<R_e) — old reduction cross-check, cleaned + He I + M10 sky masks	262.72 ± 17.99 km/s (asymmetric −18.10 / +17.88)	`scripts/run_wide_sigma_e.py --cube headline_clean_hei`, caches at `results/run_wide_sigma_e/headline_clean_hei/`. The 6.90 km/s Δ to the cleaned + He-I + M10-sky-masked new cube is the source of the refined ±3.45 km/s reduction-pass systematic
σ_e(<R_e) — pre-M10 cross-check (cleaned + He I, NEW cube)	271.87 −17.99/+17.74 km/s	Pre-M10 reference. +2.25 km/s shift to current headline = bias from un-masked OH/sky residuals folded into the M10 systematic
σ_e(<R_e) — pre-He-I cleaned cross-check (NEW cube)	268.98 −18.19/+17.98 km/s	`--cube new_clean`. +2.89 km/s offset (un-masked He I 3819 contamination biased σ_e *down*)
σ_e(<R_e) — pre-He-I cleaned cross-check (OLD cube)	260.44 −18.36/+17.97 km/s	`--cube headline_clean`
σ_e(<R_e) — legacy un-cleaned wide-window cross-check (NEW cube)	265.76 ± 17.87 km/s	Pre-clean reference; `scripts/run_wide_sigma_e.py --cube new`, caches at `results/run_wide_sigma_e/new/`

Quantity	Value	Source / notebook
σ_e(<R_e) — legacy un-cleaned wide-window cross-check (OLD cube)	254.85 ± 17.87 km/s	Pre-clean reference; `scripts/run_wide_sigma_e.py --cube headline`, caches at `results/run_wide_sigma_e/headline/`
σ_e(<R_e) — narrow Ca H+K+G cross-check window (legacy old cube, no clean)	267.95 ± 30.10 km/s	nb09d (`results/sigma_e_final_systematics_nb09d.npz`)
σ_e(<R_e/2) — gradient diagnostic	~225 km/s (NARROW)	nb07c §6cum
R_e (paper headline)	2.097″ = 15.26 kpc	best-mask F140W + F200LP CoG; `scripts/final_sigma_e.py:curve_of_growth`
log(M★/M⊙) (paper headline, validated-fit apcorr + quiescent SED prior, 2 R_e)	11.5 ± 0.1 (stat) ± 0.2 (sys) (2dp 11.46)	Bagpipes; `mstar_headline_quiescent.py` (2026-06-16 audit; was 11.50)
log(M★/M⊙) Sersic-total cross-check	11.49	Bagpipes refit, quiescent prior
z_deflector	0.67564	nb04 multi-line Gaussian fit (NIST air rest λ)
z_source	1.302	AGEL DR2, independently cross-checked via Fe II UV multiplet in KCRM blue arm (I.2.2)

The headline σ_e and σ_e cross-check are consistent at 0.4σ (13.1 km/s difference, both budgets cover it).

Part I — Spectroscopic analysis (σ_e)

I.1 KCWI integral-field spectroscopy

Cube actually used for σ_e (KCRM red arm): `Nov17_2025_DESJ0206_RL_combined_icubes_wcs.fits` (265 MB, shape 3317 × 100 × 100, λ = 5625–8941 Å, Δλ = 1.0 Å/pix, 0.300″ spaxels, ~30″×30″ FoV). Symlinked at the repo root from `../velocity_dispersion_from_IFU/`.

Complementary blue-arm reduction: `raw_KCWI/blue/DESJ0206_medium_combined.fits` (299 MB, shape 2595 × 120 × 120, λ = 3278–5872 Å, Δλ = 1.0 Å/pix, 0.300″ spaxels, ~36″×36″ FoV). Same nights as the red cube, KCRM Blue-Low grating. Not used for σ_e (the deflector absorption features that anchor σ_e are in the red), but used for the source-redshift cross-check in I.2.1 below — at z_source = 1.302 the blue cube covers source-rest ~1424–2551 Å, which contains the Fe II UV resonance lines.

Provenance correction — verified against the Keck observing logs (2026-05-26). The cube's FITS header (`DATE-OBS = 2025-11-17, PROGNAME = U002, PROGPI = Jones, DATE-BEG = 2025-11-17T09:02:15`) describes only the FIRST input frame of the most recent contributing night (`kr251117_00129`, the first DESJ0206 RED frame on Nov 17), not the full input set. The cube comment *"Reduced and stacked by Kaustubh Rajesh Gupta at UTC 2026-01-07 00:39:38"* records the actual stacking date and reducer. The science data is a multi-night combine of two confirmed AGEL collaboration nights plus a possible third:

Night (UT)	Program	PI	DESJ0206 frames	Source
2025 Aug 30	K409	TBD	12 RED `kr250830_00090-00101` (60 min) + 4 BLUE `kb250830_00052-00055` (66 min)	local `raw_KCWI/provenance/K409_2025-08-30_UTC.html` (Drive id `1yoH_elZqEsFHPRc6f-XCB98l-7dGJewN`)
2025 Nov 17	U002	Jones	20 RED `kr251117_00129-00148` (90 min) + 5 BLUE `kb251117_00087-00091` (110 min)	Drive `NightLog_KCWI_2025-11-17.pdf` (id `1HVNoH2CSAc214b_PieBQ_UkeEasVl6q4`)

Night (UT)	Program	PI	DESJ0206 frames	Source
2024 Dec 29	TBD	TBD	4 RED `kr241229_00092-00095` per local `dec29_2024_rl8000.list` — NOT independently confirmed as DESJ0206 pointings; no machine-readable observing log exists in Drive (only image-only PDF `16zWO8yaCIvRtoCGenNTndvB7dYdrCkMH`)	(raw frames on Yuguang Chen's machine; alignment QA PNGs in Drive id `1ELcTQiXV9m04Y8sA9st3OlpsNk7LEOD7`)

Total confirmed on-source integration: ≈ 170 min RED + 176 min BLUE over Aug 30 + Nov 17. RL grating central wavelengths differ between nights (Aug 30: 7400 Å; Nov 17: 7150 Å) — both rebinned onto a common 1.0 Å/pix grid spanning 5625–8941 Å by the kcwiRedux pipeline.

The user (Cordova) is on the observer list for both Aug 30 (K409 crew: Alcorn, Cordova, Glazebrook, Gonzalez-Lopezlira, Jones, Kacprzak, Tran, Vasan G C) and Nov 17 (U002 crew: Alcorn, Barone, Chen, Cordova, Glazebrook, Gupta=Kaustubh, Jones, Kacprzak, Rhoades, Tran, Vasan G C). A previously-recorded "Sept 29 K409 contribution" is incorrect — the local Sept 29 log shows zero DESJ0206 entries. For the paper, cite the two confirmed nights (Aug 30 K409 + Nov 17 U002) and flag the Dec 29 frames as a possible third-night contribution pending header verification on the raw FITS files. See `reference_kcwi_data_properties.md` and A1.

Instrumental line-spread function (B1)

`DISPSCAL = 0.2940` (instrumental σ in pixels) × `CD3_3 = 1.000 Å/pix` → FWHM_inst = 0.692 Å (constant in observed Å). Across the fit window the instrumental dispersion is σ_v,inst ≈ 12.6 km/s observed (R ≈ 10000), or 7.5 km/s rest at z_def = 0.67564. The deflector velocity dispersion (~270 km/s) is therefore resolved by a factor of ~20.

The instrumental LSF is passed to ppxf as a per-pixel rest-frame FWHM dict (`{"lam": lam_gal_rest, "fwhm": fwhm_gal_rest}` at `scripts/bootstrap_ppxf.py:_prep_spectrum_for_ppxf`), the canonical Cappellari (2023) interface. `sps_lib` interpolates this onto each template's native wavelength grid and pre-broadens via `util.varsmooth`. We verified (B2) that scaling DISPSCAL across the range {×0.5, ×0.75, ×1.0, ×1.25, ×1.5, ×2.0} shifts σ by at most 0.83 km/s — most templates are intrinsically broader than the KCWI LSF, so ppxf's `fwhm_diff² = (FWHM_gal² − FWHM_tem²).clip(0)` clamp at `sps_util.py:169` means no convolution is applied for FSPS or EMILES, and only sub-km/s deconvolution for XSL. The LSF subtraction is rock-solid relative to the headline precision.

Seeing

Guide-camera FWHM = 1.27″ (`GUIDFWHM` header from the Nov 17 stacking pass). Used as a conservative seeing estimate; the K409 Aug 30 log gives airmass 1.07–1.08 but the per-night DIMM seeing has not yet been extracted from the observing-log header (open item flagged in `reference_kcwi_data_properties.md`). The R<R_e/8 aperture (0.288″) was dropped because it sits inside the seeing FWHM/2 = 0.64″ (D5).

I.2 Redshifts

I.2.1 Deflector systemic redshift (A2)

The deflector redshift was redetermined by independent Gaussian fitting of seven absorption / emission lines in the integrated spectrum, anchored on NIST air rest wavelengths (Ca K 3933.66, Ca H 3968.47, G-band 4304.40, Hδ, Hγ, Hβ, [OII] doublet with shared parameters). The result, z_systemic = 0.67564, supersedes the AGEL DR2 catalog value 0.67511 (which is offset by ~95 km/s — used only in the older notebooks 03/03b). Notebook 04; module `scripts/redshift_verify.py` provides the `ABSORPTION_LINES`, `EMISSION_LINES`, and fitting routines.

All wavelength bookkeeping downstream uses NIST air rest wavelengths; see I.5 for how the air↔vacuum convention is handled between the galaxy spectrum and the SPS template libraries.

I.2.2 Source redshift cross-check via Fe II UV absorption (blue arm)

The AGEL DR2 source redshift z_source = 1.302 was independently cross-checked using the KCRM blue-arm cube (`raw_KCWI/blue/DESJ0206_medium_combined.fits`, λ_obs = 3278–5872 Å, see I.1). At z = 1.302 this observed-frame range covers source-rest ~1424–2551 Å, containing the canonical Fe II UV resonance multiplets that are the standard low-ionization absorption-line diagnostic in intermediate-z galaxy ISM spectra. The Fe II UV3 / UV2 lines fall cleanly inside the blue cube; the strong UV1 doublet (Fe II λ2587, λ2600) lies marginally past the blue-arm red edge at 5872 Å (would appear at obs ~5957 / 5988 Å):

Source-rest line	Source-rest λ (Å, air)	Observed λ at z=1.302 (Å)	In blue cube?
Fe II UV3	2344.21	5397.4	✓
Fe II UV2	2374.46	5466.9	✓
Fe II UV2	2382.76	5486.0	✓
Fe II UV1	2586.65	5955.5	(just past red edge)
Fe II UV1	2600.17	5986.6	(just past red edge)

These features, fit jointly on the arc spectrum extracted from the blue cube, recover a source redshift consistent with the AGEL DR2 value z_source = 1.302. [TBD: insert measured z_source ± uncertainty, notebook/script reference, and result-cache path once the analysis is committed — the work was done on the blue arm but the artifact is not yet in the repo as of 2026-05-19.]

The Fe II UV multiplet identification is the cleanest available verification for z_source in this dataset: the source is a star-forming spiral (so the resonance Fe II absorption is expected to be present); all five lines have published air-rest wavelengths to ≪ 0.01 Å (NIST); and joint fitting with shared (z, σ) collapses the per-line statistical uncertainty by $\sqrt{N}$. The Mg II λ2796, λ2803 doublet from the same source (red cube, def-rest 3835–3855 Å) is independently the strongest detection used for the source-emission spectral mask (I.4), giving a second handle on z_source via the same physical absorber.

I.3 Aperture and I-weight construction

The headline σ_e is measured on a single I-weighted, 1-D aperture spectrum extracted at R<R_e from the IFU cube. This matches the literature standard (Cappellari et al. 2006, SAURON IV §2.3 co-add-then-fit; σ_e integral = Gültekin 2009 eq. 1 — NOT Cappellari 2006 eq. 1, which is the aperture correction; as adopted by Kormendy & Ho 2013, Greene et al. 2020 ARA&A, SAURON/ATLAS3D, MaNGA).

Center and aperture radius

- Center. F140W and F200LP cutouts are centroided with `photutils.centroid_2dg` on the deflector core (each with its own `_mask.fits`), averaged in world coordinates, then propagated to IFU sub-pixel via the KCWI WCS (`scripts/final_sigma_e.py:find_center`). Adopted: RA = 31.55613°, Dec = −1.23819° (ICRS). The F140W vs F200LP centroid offset is 0.36″ — driven by the F200LP arc dragging that filter's centroid; F140W is the cleaner bulge centroid, and the HST-mean is the robust compromise. The 5-center sweep (F1) gives σ_e spread = 3.7 km/s, carried as the ±4 km/s "centering" component of the systematic budget.
- Aperture radius. R_e = 2.305″ = 16.23 kpc, the mean of the F140W and F200LP masked curves-of-growth (see II.3). Three apertures were considered: R<R_e/8, R<R_e/2, R<R_e (D5). R<R_e/8 was dropped because at 0.288″ it sits inside seeing FWHM/2; the headline is at R<R_e (KH13 / Greene+20 σ_e definition).

Spatial arc mask (E1)

The lensed source at z=1.302 contributes flux at several IFU spaxels along the arc that pass into the R<R_e aperture as low-velocity continuum. We mask them using the F200LP segmentation mask (`AGEL020613-011417A_F200LP_WFC3_cutout_L3_mask.fits`, 500 × 500 at HST 0.05″/pix). This mask is arc-only by construction; the F140W `_mask.fits` is broader and also covers the deflector core, so it is *not* an arc mask and never used as one (see `feedback_masking_strategy.md`, A4/I2 in TESTS_AND_DIAGNOSTICS).

The HST mask is reprojected to the IFU 0.300″ grid via `scipy.ndimage.map_coordinates(order=0)` (nearest-neighbor). Inside R<R_e: 38 of 184 spaxels are flagged (~21% by count, ~27% by I-weight).

I-weight construction

For each spaxel inside R<R_e (and unflagged by the spatial mask), the flux weight is the per-spaxel collapse of the IFU 6500–7500 Å white- light band (the same window used in the narrow-window ppxf fit, so the weighting and the fit have matched footprints). The weighted 1-D spectrum is summed (not averaged) so the wild bootstrap can preserve the per-spaxel noise contribution.

The choice of I-weight map is one of the systematic axes (D3, J6). We bracket it with the 10-shape sweep (`scripts/run_isource_shape_sweep.py`): IFU_band (headline), IFU white-light, F140W and F200LP × {raw, arcmasked, 1D-CoG smooth, 2D-Sersic smooth}. After the 2026-05-11 Sersic2D bound-fix (J5; see I.5 below) the I-shape spread is ±1.5 km/s at the wide arc-masked window (was ±13 km/s at narrow before the bound-fix removed the F200LP n=0.30 pathology).

I.4 Wavelength fit window

This is the methodological pivot of the paper: there are two windows in play, and the headline moved to the wider one during May 2026.

Narrow Ca H+K + G window (cross-check)

`w6500_7500` = obs 6500–7500 Å = rest 3879–4476 Å at z = 0.67564. Anchored on Ca H+K and the G-band. This is the window used by the older nb07c §6cum pipeline (now retained as the method cross-check). 927 good pixels. σ_e(<R_e) = 267.95 ± 30.10 km/s.

Wide arc-masked window (current paper headline)

`wR3800_5400_arcmask` = rest 3800–5336 Å, with explicit masks at four deflector-rest bands that are contaminated by emission from the z=1.302 source (factor (1 + z_s)/(1 + z_l) = 1.374 maps each source-rest line into the deflector rest frame). 2161 good pixels — a 2.6× increase in spectral information vs the narrow window. σ_e(<R_e) = 254.85 ± 17.87 km/s.

Spectral-mask catalog (J3, codified at **`scripts/run_window_sweep.py:ARC_MASKS_REST`**): three source-emission masks plus one telluric.

Identification	Source-rest λ (Å)	Deflector-rest λ (Å)	Obs λ (Å)	Masked band, def-rest (Å)
Mg II λ2796, λ2803 (z=1.302)	2796, 2803	3842, 3852	6438, 6455	3835–3855
O_2 A-band leading-edge telluric	—	—	7593–7626	4525–4545
[O II] λ3727, λ3729 (z=1.302)	3727, 3729	5121, 5124	8581, 8585	5115–5135
[Ne III] λ3869 + Mg b cluster (z=1.302)	3869	5314	8903	5260–5340

Total: 212 of 2374 pixels (8.9% of the fit window) excluded. The catalog was identified empirically (>4σ excesses above local continuum in the wild-bootstrap posterior, J2). Mg II, [O II], and [Ne III] are physically identified at z = 1.302. The fourth band, originally tagged "source rest ~3300 Å unidentified," was confirmed on 2026-05-18 to be a telluric residual at the leading edge of the atmospheric O_2 A-band (obs 7593–7626 Å): the spike sits at the same observed wavelength in the deflector aperture, the CLAUDE.md sky box, and a 4–8″ off-deflector annulus, with deflector/off-source amplitude ratio 1.11× despite a 17.5× continuum-brightness ratio — the unambiguous signature of an additive wavelength-locked systematic, not a localized source line. See `NOTES_4534A_spike_investigation_2026-05-18.md` and `results/figures/spike_4534A_diagnostic.png`. The mask was placed for the right reason (kills the spike) but for the wrong attribution; the mask itself is unchanged.

This catalog is the first-of-its-kind methodology for AGEL deflector kinematics. It is reusable for any AGEL target where z_source is known.

Why the wide window is the headline (not just incrementally better). Three independent improvements stack at the wide arc-masked window (documented in nb09d and figured in `results/figures/nb09d_per_sps_both_windows.png`):

1. 4× tighter statistical error. 2161 good pixels vs 927; the wild-bootstrap stat 1σ shrinks from ±23.9 to ±6.1 km/s (J6).
2. SPS-template spread collapses 26 → 4 km/s. At the narrow window the three SPS libraries disagree by 26 km/s with strong ordering (FSPS=253.7 < EMILES=267.9 < XSL=279.6; see also `reference_sps_systematic.md`). At the wide arc-masked window they agree to 4.2 km/s (FSPS=253.9, EMILES=253.3, XSL=257.5) (J4). Mg b + Fe5270 + Fe5335 + the broader feature set anchor the templates well enough that the FSPS/EMILES/XSL ordering vanishes.
3. F200 spatial-mask sensitivity drops 4×. From ±7.5 km/s (narrow) to ±3.8 km/s (wide) (J7) — the spectral arc-emission masks absorb most of what the F200 spatial mask used to absorb; the spatial mask becomes a sub-budget perturbation.

Provenance of the wavelength range. Not from a specific literature prescription. The 3800 Å blue edge is set by the XSL template floor at def-rest ~3675 Å plus a 5% velocity padding; the 5336 Å red edge is set by KCWI red-channel coverage at z = 0.67564. The closest literature precedent for going as blue as 3800 Å is LEGA-C (Bezanson et al. 2018; van der Wel et al. 2021) at z ~ 0.7 using ~3500–5500 rest. SAURON/ATLAS3D's canonical kinematic window is narrower (4800–5380; Mg b + Fe5270 + Fe5335 only). The Lick-tradition "rest 4000–5400 (no Ca H+K)" range is captured by the `wR4000_5400_arcmask` sensitivity check (see I.6 below), not the headline. A drafted statement claiming "Cappellari 2017 recommends extending blueward" was a hallucination — retracted.

I.5 ppxf setup

We use `ppxf` v9.x (Cappellari & Emsellem 2004; Cappellari 2017; 2023) with three SPS template libraries pooled at the posterior level.

SPS libraries and the vacuum/air fix (B3, C1–C4)

All three libraries shipped with ppxf are used and pooled:

Library	File	Native wavelength frame	V_sys diagnostic
FSPS	`spectra_fsps_9.0.npz`	vacuum	Ca K minimum at 3934.86 Å (+1.20 Å vs air rest)
EMILES	`spectra_emiles_9.0.npz`	air	Ca K minimum at 3933.82 Å (+0.16 Å)
XSL	`spectra_xsl_9.0.npz`	air	end-to-end V_sys closure at +9 km/s with air galaxy

The original pipeline (nb07c era) converted the galaxy to air for *all three* libraries. Air-galaxy ↔ vacuum-FSPS produced a spurious −90 km/s FSPS V_sys offset (matches the air↔vacuum differential at 6500–7500 Å, −83 km/s per Ciddor 1996). The fix (`scripts/bootstrap_ppxf.py:SPS_NATIVE_FRAME` + `frame_galaxy='auto'`): keep the galaxy in vacuum for FSPS, apply the scalar-median `util.vac_to_air` ratio for EMILES and XSL. After the fix the V_sys split-track collapses from ~110 km/s to ~15 km/s and the σ shift on the 3-SPS pool is ≤ 2 km/s

(C2, C3). The full derivation is in `NOTES_nb09_frame_fix_and_final_sigma_e_2026-04-28.md`, including the canonical Cappellari pattern citation (`ppxf_example_kinematics_sdss.py`:127-134) and Ciddor (1996) for the conversion accuracy.

We follow Cappellari's pattern of applying `vac_to_air` at observed wavelengths (rather than rest); the alternative would shift V_sys by 1.83 km/s at our z (B5), sub-budget.

Per-SPS V_sys subtraction before pooling (C4)

The σ posteriors from each SPS are concatenated raw (σ is invariant to V offsets), but V_sys medians are subtracted per-SPS before pooling so that residual template-systematic offsets (now FSPS −19, EMILES −4, XSL −1 km/s post-frame-fix) don't inflate the V posterior. The pooled σ posterior is the 16/50/84 percentile of the concatenated all-3-SPS sample (typically $\geq 10^4$ samples after 30 polynomial degrees × N=500 bootstrap × 3 SPS).

Polynomial degree (G3)

Multiplicative polynomial degrees 15–29 (15 values) are stacked into the per-SPS posterior. The σ–vs–degree diagnostic (`results/figures/nb09_sigma_vs_degree.png`) is flat within the bootstrap envelope across this range — the polynomial is saturated and no monotonic trend (which would indicate the polynomial absorbing LOSVD signal) is present. The lower bound (deg=15) is set by the requirement to flatten the global continuum across the wider arc-masked window; deg<15 leaves systematic residuals.

Wild bootstrap (G1, G2)

For each (SPS × degree) we run a hybrid Rademacher × local-residual wild bootstrap (`scripts/bootstrap_ppxf.py:run_bootstrap_single_degree`). Per iteration: residual = (galaxy − bestfit) × Rademacher sign, locally scaled by σ from a 75-pixel rolling window, re-fit, record V and σ. Production statistics is N=500 per (SPS × degree); N=50 smoke runs are used for development. N=50 vs N=500 agree on σ within 1 km/s (G4). Parallelization uses joblib with BLAS pinned to 1 thread per worker (`scripts/bootstrap_ppxf_parallel.py`) — without the pin, BLAS thread oversubscription on Mac multicore caused a >4 h pathology on one fit.

I.6 σ_e measurement

The pipeline applied per window:

```
1. Extract the I-weighted aperture spectrum at R<R_e (I.3).
2. For each SPS:
     a. Set the galaxy wavelength frame to match the SPS native frame (I.5).
     b. For each polynomial degree d ∈ {15, …, 29}:
          Run wild bootstrap × N=500.
3. Subtract per-SPS V_sys median; concatenate σ samples across SPS × degrees.
4. Headline σ_e = median of the pooled posterior; stat 1σ = pooled (84−16)/2.
```

The headline σ_e values are reported in §0. Full per-SPS, per-aperture, per-polynomial-degree posteriors are cached in `results/nb09a_wavelength_sweep/{window}_{sps}_T*_N500.npz` for the wide window, and `results/final_sigma_e_paper/` for the narrow window.

Three-window cross-check (J8)

A third window beyond the two headlines, `wR4000_5400_arcmask` (rest 4000–5400, no Ca H+K), was run at full N=500 as an orthogonal feature-set check (Hβ + Mg b only, the Lick-tradition window). Its σ agrees with both headline windows within the budget:

Window	rest pixels	σ_e (3-SPS pool, N=500)	role
w6500_7500	3879–4476	269.7 km/s	NARROW cross-check

Window	rest pixels	σ_e (3-SPS pool, N=500)	role
wR3800_5400_arcmask	3800–5336	254.8 km/s	HEADLINE
wR4000_5400_arcmask	4000–5400	~265 km/s	orthogonal Hβ+Mg b check

The 15 km/s spread among these three windows defines the fit-window systematic carried in both budgets (now the dominant residual).

I.7 Independent cross-check estimators

Three architecturally-independent σ_e estimators were built and compared at the narrow window (none share intermediate state with the cumulative I-weighted aperture fit). They all agree within budget; see `reference_cumulative_vs_annular_sigma_e.md` for the design rationale, `reference_sigma_e_estimator_independence.md` for the independence guarantee.

Estimator	Method	σ_e(<R_e) NARROW	vs §6cum
§6cum (paper cross-check)	Cumulative I-weighted aperture ppxf	267.32 ± 24 km/s	—
§7 (annular Gültekin)	Discrete annular sum of $F\cdot(V^2+\sigma^2)$ (H1)	256.17 −13.0/+12.7 km/s	−11 (<1σ)
§7b (flat-σ extrapolation)	§7 + outer-annulus σ extrapolation (H2)	274.37 −16.2/+17.4 km/s	+7 (<1σ)
nb07e (arc subtraction)	§6cum with α·arc subtracted as additive template (E7)	matches §6cum within 0.1 km/s	—
nb07f (arc-as-sky)	§6cum no-mask + arc as ppxf `sky=` template (E8)	274.64 (recovery=145%)	confirms dilution mechanism

§7 (discrete annular sum) is the σ_e definition often associated with Gültekin et al. (2009); we use it for the radial σ(R) profile and the per-annulus systematics inspection. The §6cum cumulative I-weighted aperture is the literature-standard σ_e definition for M•–σ work (Cappellari et al. 2006, SAURON IV §2.3 co-add-then-fit; σ_e def. = Gültekin 2009 eq. 1, also used by KH13 / Greene+20 / SAURON / ATLAS3D / MaNGA), and is the cross-check the paper headline is calibrated against. The arc-as-sky test (E8) is the rigorous test of the no-mask Δ mechanism: with the arc spectrum as a free-amplitude additive template, σ recovers to 274.6 km/s, +6.8 km/s above the masked headline — the 145% recovery fraction confirms that the no-mask bias is entirely continuum dilution and there is no kinematic-blend mechanism contributing in the opposite direction. The 7 km/s overshoot is consistent with a small amount of real deflector light leaking into the outer-arc-mask spaxels that get oversubtracted.

I.8 Resolved kinematics (supplementary)

The wide arc-masked window also enables per-spaxel and PowerBin kinematic maps that failed at the narrow window (Test 19 in nb05x). Documented in `notebooks/11_perbin_perspaxel_kinematics_wide.ipynb` and `project_nb11_resolved_kinematics.md`.

- Per-spaxel at S/N ≥ 5 (K1): 17 central spaxels inside R<1.5 R_e with σ ∈ [144, 251] km/s, zero outliers above 400 km/s, median σ = 201 km/s. σ(R) drops with radius — consistent with the elliptical-bulge expectation (see `feedback_deflector_morphology.md`).
- PowerBin at target S/N=15 (K2): 7 bins inside R<1.5 R_e, median σ = 294 km/s. 5 of 7 inner bins are clean; the 2 outer bins still hit σ > 400 km/s (KCWI per-spaxel S/N limit at the aperture edge, NOT a binning failure).
- K3: at the narrow window neither approach worked — outer bins σ > 800, no per-spaxel S/N above floor. The wide arc-masked window delivered a 4× increase in per-pixel S/N that unlocked the maps.

These are supplementary figures, not the headline.

I.9 Spectroscopic systematic budget — both windows

The headline σ_e is the median of the pooled posterior on the NEW `_mtwdo_` reduction; the error is the quadrature sum of seven independent components (one more than the original budget — the "reduction-pass" component added 2026-05-26, refined again 2026-05-27 when the He I 3819 source-emission mask was added):

Component	Wide arc-masked, NEW `_mtwdo_` + clean + He I + M10 + M11 (headline)	Source
Statistical (pooled 1σ)	±4.6 (asym −5.16/+4.13)	wild bootstrap N=500 × 3 SPS × 15 degrees on cleaned + He-I + M10-masked NEW cube
I-shape (10-shape spread, M11)	±2.27	`results/ishape_sweep_wR3800_5400_arcmask_M10/` — peak-to-peak/2 of 10 I-shape posteriors (range 266.83–271.37 km/s)
F200 spatial mask (peak-to-peak/2, M11)	±6.65	`results/maskweight_sweep_wR3800_5400_arcmask_M10/` — (w00=269.69, w50=261.95, w100=256.39)/2 = 6.65
Frame (vac/air per SPS, structural)	±5.0	`audit_ppxf_methodology.py` audit 1
Centering (5-center sweep ±0.4″, carried)	±4.0	`NOTES_centering_investigation_2026-04-27.md`
Fit-window (3-window spread, M11)	±3.82	`results/nb09a_wavelength_sweep_M10/` — wR3800_5400_arcmask 269.66, wR4000_5400_arcmask 268.58, w6500_7500 276.22 → peak-to-peak/2 = 3.82. Dramatic drop from carried ±15 because the cleaned NEW cube has wide/narrow agreement (vs ±15 spread on OLD cube)
Reduction-pass (M10)	±3.45	half-Δ between cleaned + He-I + M10-sky-masked new and old reductions = (269.62−262.72)/2 = 3.45 km/s
Quadrature total (symmetric)	±11.77	$\mathrm{sqrt}(2.27^2 + 6.65^2 + 5.0^2 + 4.0^2 + 3.82^2 + 3.45^2)$
Asymmetric total (lower / upper)	−11.98 / +11.57	preserves stat-side skew

2026-05-27 M11 rigorous re-derivation result: total sys dropped from ±17.78 → ±11.77 km/s (down 6.0). The fit-window component dropping from ±15 → ±3.82 is the dominant change. The carried ±15 was from pre-clean OLD-cube nb09 §9; the cleaned NEW cube + M10 masks bring the three windows into much closer agreement.

OLD cube cross-check at the same M10 mask configuration (carried components still apply for it, since it's not the headline): σ_e = 262.72 ± 17.99 km/s. Narrow Ca H+K+G cross-check on OLD cube (legacy): σ_e = 267.95 ± 30.10 km/s.

The fit-window systematic (±15 km/s) is still the dominant residual. The other six components quadrature-sum to ~10.5 km/s. Tightening the window systematic would require additional N=500 windows at different feature subsets (Hβ-only, Mg b-only) — not cheap, and only marginally useful given the dominant role of the 13 km/s wide-vs-narrow shift.

On using "old" component values for the new headline: the I-shape, F200-mask, and fit-window systematics were measured on the OLD cube. We assume they scale similarly on the NEW cube (component-magnitude phenomena from the same input frames + same arc + same I-weight maps, just different flat-fielding) and carry the OLD values unchanged. Strict re-derivation on the new cube would cost ~50 min N=500 × 3 SPS × 10 shapes + 3 mask weights — possible but expected to give the same ~1-km/s-level numbers since the underlying frames and aperture are unchanged. Flagged for re-derivation if needed.

Note on independence. The budget components are not strictly statistically independent: I-shape is bounded by mask choice; frame is bounded by SPS choice; centering is partially absorbed by the multiplicative polynomial. Quadrature combination is therefore *conservative* — the true total error could be ~10% smaller. We report the conservative number.

Cross-references for individual budget items

- Statistical: bootstrap convergence at N=500 (G4); polynomial saturation (G3); per-SPS V_sys subtraction before pooling (C4).
- I-shape: 10-shape sweep (D3, D4, J6); Sersic2D bound-fix prevents the n=0.30 pathology (J5, applied 2026-05-11 at `scripts/run_isource_shape_sweep.py:181-207`).
- F200 mask: hard/no-mask Δ (E2, E3); 5-point mask-weight sweep with quadratic c = +7.3 (threshold-dominated bias, E5); arc dilation rejected as inflating σ via noise (E6, negative finding); arc-subtraction sibling matches §6cum (E7); arc-as-sky 145% recovery confirms dilution mechanism (E8).
- Frame: per-SPS frame identification + V_sys closure (B3, C1, C2); σ shift sub-budget (C3).
- Centering: HST-mean centroid_2dg (A4, F2); 5-center sweep (F1).
- Fit-window: nb09 §9 three-window spread (J8); window sweep underpinning (J1); source-emission catalog (J2, J3).

What is *not* in the budget (intentionally)

- R_e source. Four R_e definitions (mean, F140W-only, F200LP-only, Ca H+K+G depth-map) give σ_e at the narrow window with 16.9 km/s spread (D7) — at the mask-budget level, but sub-budget vs the total ±30. σ_e rises monotonically with R_e (more outer-bulge spaxels in the I-weighted aperture). This is treated as a method sensitivity rather than a budget addition.

- Reduction-pass (M1) — PROMOTED to the formal budget (2026-05-26). The new `_mtwdo_` reduction (with hybrid master-twilight + dome flats) is now the headline σ_e source; the OLD reduction is retained as a second-reduction cross-check. The half-Δ between the two reductions (= 5.46 km/s) is carried as the "reduction-pass" component of the systematic budget (see I.9 table above). Flag: only two reductions available — revisit and refine if a 3rd reduction lands. Audit: `NOTES_nb09e_audit_2026-05-20.md` + memory `project_nb09e_reduction_systematic.md`; cross-test for localized mechanism: rejected (the +10.9 km/s shift is not from a single wavelength boundary — see Addendum 2026-05-26 (b) in the project memory).

- Local-MAD bad-pixel cleaning (M5, 2026-05-26 sensitivity, NOT in budget). ppxf `clean=True` rejects 0 pixels on this dataset because the noise array is overestimated relative to the actual residual scatter. Using local-MAD-based outlier detection (rolling 75-pixel window, |residual|/ local_MAD > 3σ) on the canonical residuals identifies 52 cosmic-ray- like spikes that the noise-scaled clean=True missed. The biggest is a 6-pixel cluster at def-rest 5232–5236 Å (obs 8768–8774 Å) with a 26σ amplitude in local-MAD units — a clear unresolved CR residual. Clipping these 52 pixels and re-running at N=100 shifts σ_e UP by:

  - NEW cube: 265.76 → 267.18 km/s (Δ = +1.42)
  - OLD cube: 254.85 → 256.01 km/s (Δ = +1.16)

  Both cubes shift by the same amount → the 52 bad pixels are intrinsic to the data (CR residuals shared by both reductions), NOT reduction-specific (M6, 2026-05-26 replication). The +10.9 km/s reduction-pass gap is preserved (10.91 un-clipped → 11.17 clipped). An alternative noise-scaled approach (scale noise array by 0.3 and re-run ppxf clean=True) finds only 19 pixels per fit but gives a larger σ_e shift (+5.28 km/s to 271.04) — different pixel-selection algorithm; the local-MAD result is more defensible because the same flagged pixels replicate the cleaning shift on the OLD cube.

  We carry this as a stated sensitivity rather than folding into the formal budget because (a) the shift is well below stat 1σ (±6.1),

  (b) cleaning preserves the reduction-pass gap, (c) the choice of threshold (3σ local-MAD) is not strongly motivated vs alternatives. Caches: `results/run_wide_sigma_e/new_local_mad_clip_N100/`, `results/run_wide_sigma_e/old_local_mad_clip_N100/`, and `results/run_wide_sigma_e/new_noise_scaled_clean_N100/`. Scripts: `/tmp/test_local_mad_clip.py`, `/tmp/test_local_mad_clip_OLD_cube.py`, `/tmp/test_noise_scaled_clean.py`.

- R<R_e/2 vs R<R_e aperture choice. Aperture is a methodological choice, not a systematic. R<R_e is the KH13/Greene+20 convention and the paper headline; R<R_e/2 is reported as a gradient diagnostic.

- σ–vs–degree trend. Flat across deg 15–29 (G3); no contribution to budget. If a future referee asks, the trend is shown inline in nb09 §6.5.

Part II — Photometric analysis (R_e, M★)

II.1 Imaging data

Four bands enter the SED fit and the morphological measurements; full provenance in `reference_hst_jwst_data_properties.md`.

Telescope	Instrument	Filter	Program	PI	Date (UT)	Drizzled scale	Exptime	ZP_AB
HST	WFC3/UVIS	F200LP	16773	Glazebrook	2022-07-14	0.05000″/pix	600.0 s	27.344
HST	WFC3/IR	F140W	16773	Glazebrook	2022-07-14	0.08000″/pix	597.7 s	26.446
JWST	NIRCam SW (Mod B)	F150W2	05594 (Cluster SLICE)	Mahler	2024-08-27	0.03075″/pix	1836 s	28.033
JWST	NIRCam LW (Mod B)	F322W2	05594	Mahler	2024-08-27	0.06301″/pix	1836 s	26.475

AB zeropoints are derived per image from the FITS headers (the standard HST formulation `ZP_AB = −2.5·log10(PHOTFLAM) − 21.10 − 5·log10(PHOTPLAM) + 18.6921`; JWST `ZP_AB = −6.10 − 2.5·log10(PIXAR_SR)` on the MJy/sr i2d products). Zeropoints are read from headers; never hardcoded.

Plate scale at z_def = 0.67564 (FlatLambdaCDM, H_0=70, Ω_m=0.3): 1″ ≈ 7.04 kpc. At the four drizzled scales above this is 352, 564, 217, and 444 pc/pix respectively.

Drizzle pixel correlation. Noise in drizzled images is correlated between adjacent pixels — empirical noise is ~1.3–2× higher than the idealized CCD-equation prediction. The STScI ETC does not account for this. Relevant for aperture photometry error bars; we propagate via the per-pixel noise map produced by the drizzle pipeline rather than recomputing from photon statistics.

II.2 Centering

Identical to I.3 — `photutils.centroid_2dg` on F140W and F200LP cutouts, averaged in world coordinates. RA = 31.55613°, Dec = −1.23819° (ICRS). The F140W/F200LP centroid offset is 0.36″ (driven by the F200LP arc contamination); the HST-mean is the robust adopted center.

II.3 Effective radius (A3)

R_e = 2.305″ = 16.23 kpc, the mean of the F140W and F200LP masked curves-of-growth (`scripts/final_sigma_e.py:curve_of_growth`). The masked CoG zeros mask-flagged HST pixels before the radial flux integral, so the lensed arc and any companion contributions do not bias the bright-end half-light point.

Band	Masked CoG R_e (″)
F140W	2.168
F200LP	2.441
Mean (headline)	2.305

Production vs measurement script — important not to confuse them. The production R_e (2.305″) comes from `final_sigma_e.py:curve_of_growth` using HST-mean `centroid_2dg` centers. A separate older script scripts/measure_Re.py (results/Re_measurements.npz) uses the cutout image-geometric center and produces 10 variants under different masking strategies; its "proper mask" rows give a different mean (2.633″). Cite the production value.

The earlier IFU white-light value ($R\_e = 2.61''$, used in nb05/06 era) is superseded by the HST-based $R\_e = 2.305''$, which has the F140W + F200LP arc masks built in.

R_e source as a σ_e systematic (D7). Four R_e definitions (mean, F140W-only, F200LP-only, Ca H+K+G depth-map at $R\_e = 2.866''$) shift the narrow-window σ_e by up to 14 km/s. σ_e rises monotonically with R_e — larger aperture catches more outer-bulge spaxels. This sensitivity is documented in nb09 §7b and figured at results/figures/nb09_re_source_systematic.png; it is held outside the formal budget (see I.9).

II.4 Aperture photometry

Performed interactively with scripts/photometry_masking_HST.py (HST) and scripts/photometry_masking_JWST.py (JWST) — interactive matplotlib GUI tools (click-to-mask + sliders); not headless. The HST script uses PHOTFLAM/PHOTPLAM; the JWST script uses PIXAR_SR.

The deflector aperture excludes the arc using the F200LP _mask.fits (reused from the σ_e pipeline); for the JWST bands the arc geometry is the same and re-projected from the HST WCS. The four extracted AB magnitudes are the input to Bagpipes (II.5):

Filter	λ_pivot (Å)	AB observed
F200LP	4972	22.6126
F140W	13 923	19.1335
F150W2	16 720	18.9425
F322W2	32 470	18.6042

Error bars are set to 10% fractional on each band as a conservative systematic budget (covers both the formal photon noise and the drizzle-correlation underestimate noted in II.1).

II.5 Bagpipes SED fitting (A5)

We fit the four-band SED with Bagpipes (Carnall et al. 2018, 2019). Configuration:

```
fit_instructions = {
    "redshift":   (0.674, 0.676),        # narrow spec prior, brackets z = 0.67564
    "exponential": {
        "age":           (0.1, 15.),     # Gyr
        "tau":           (0.3, 10.),     # e-folding time, Gyr
        "massformed":    (1., 15.),      # log10(M_formed / M⊙)
        "metallicity":   (0., 2.5),      # Z / Z⊙
    },
    "dust": {"type": "Calzetti", "Av": (0., 2.)},
}
```

- SPS: BPASS + CLOUDY (Bagpipes default), Calzetti dust law, free metallicity.
- Sampler: Nautilus (n_live = 400; pure-Python fallback to MultiNest if needed). 500-sample posterior.
- Cache: pipes/posterior/AGEL0206/0206_real.h5 (delete to refit).

The fit was executed end-to-end in notebooks/02_streamlined_Bagpipes_SED.ipynb (and the older exploratory 02_Bagpipes_SED_fitting.ipynb). Saved posteriors in results/bagpipes_sed_results.npz.

Posteriors

Parameter	p16	p50	p84
log(M★/M⊙) (headline)	11.24	11.33	11.41
mass-weighted age (Gyr)	3.69	5.10	6.11
exponential SFH age (Gyr)	4.46	5.98	6.97
exponential τ (Gyr)	0.43	0.68	1.21
metallicity (Z/Z⊙)	0.17	0.89	1.87
A_V (mag)	0.34	0.82	1.55
z_phot	0.674	0.675	0.676

→ M★ = 2.15×10^{11} M⊙ at the headline aperture. z_phot is foreced to agree with the line-fit z = 0.675 given the strong prior we implemented.

Per-band residuals — flagged for transparency

Filter	AB obs	AB model (p50)	Residual
F200LP	22.613	22.563	+4.7%
F140W	19.134	19.264	−11.3%
F150W2	18.942	19.033	−8.0%
F322W2	18.604	18.505	+9.6%

The model underpredicts F140W and F150W2 (the rest-frame near-IR bands) by ~8–11% and overpredicts the bluest and reddest bands by ~5–10%. This is the classic "redder NIR slope than a single-τ SFH prefers" tension — not catastrophic but flags that a more flexible SFH (delayed-τ or non-parametric) might tighten the fit if revisited. Effect on M★ is at the ±0.07–0.15 dex level already quoted.

Bagpipes plotting convention (a recurring gotcha)

`model_galaxy.wavelengths` is rest-frame Angstroms (multiply by (1+z) to plot in observed frame). `spectrum_full` flux is plotted directly — no (1+z) correction. `model_photometry` is already in observed-frame units. Confirmed against Bagpipes source at `plot_spectrum_posterior.py:69`; see `feedback_bagpipes_sed.md`.

II.6 Sersic-total photometry cross-check (A5)

Single-component 2D Sersic profiles fit per band (notebook 08; `scripts/measure_Re.py` shares the same fitting framework with the 2026-05-11 bound-fix applied — n ∈ [1.0, 6.0], ellip ∈ [0.0, 0.6], 3-init grid). Extrapolated to total flux, then re-fed through Bagpipes with identical priors:

Filter	AB aperture	AB Sersic-total	Δmag	Sersic n	r_eff (″)
F200LP	22.613	20.672	−1.94	1.42	2.49
F140W	19.134	18.282	−0.85	1.54	1.88
F150W2	18.942	18.154	−0.79	1.40	1.72
F322W2	18.604	17.633	−0.97	1.97	2.03

Quantity	Aperture (paper)	Sersic-total	Δ
log(M★/M⊙) p50	11.33 +0.07/−0.09	11.40 +0.11/−0.15	+0.065 dex (+16% in M★)

The aperture under-counts outer-profile flux by 0.8–1.9 mag depending on filter, but at the integrated-mass level the Sersic correction is +0.07 dex — within the quoted aperture-photometry uncertainties. The paper cites the aperture value as the headline; Sersic-total is the sensitivity floor.

Caches: `results/bagpipes_sed_results.npz` (aperture, 500-sample posterior), `results/bagpipes_sersic_refit.npz` (Sersic-total), `results/sersic_total_photometry.npz` (Sersic fit parameters).

II.7 Photometric systematic budget

Component	Δ log(M★/M⊙)	Notes
Bagpipes posterior (aperture, p16–p84)	+0.07 / −0.09	dominant; ~17% / 19% in M★
Aperture vs Sersic-total	+0.065 dex	sub-dominant; sensitivity floor at +16% in M★
SED residuals (NIR tension)	folded into posterior	flexible SFH might tighten ~0.05 dex; not pursued
Drizzle noise correlation	included in 10% per-band σ_flux	conservative
Centering	sub-percent (HST-mean robust)	inherited from I.3/F1

The aperture posterior ±0.08 dex is the headline uncertainty. The Sersic-total cross-check tells us the aperture is biased low by ~0.07 dex relative to a single-Sersic total — at the headline-uncertainty level, so the conservative paper choice is to quote the aperture value and note the Sersic-total as a +16% sensitivity check.

Part III — Joint summary and outstanding items

III.1 Final tabulated systematics

```
σ_e(<R_e)              = 267.31 ± 11.77 km/s  (paper headline, NEW _mtwdo_
                                                reduction, wide arc-masked window,
                                                + He I 3819 (M8) + M10 sky audit
                                                + M11 systematic re-derivation
                                                + M12 R_e-source fold-in,
                                                symmetric quadrature)
                       = 269.62 -13.45 / +13.10 km/s  (asymmetric form: stat
                                                -5.16/+4.13 from pooled
                                                bootstrap + sys ±12.43 each side;
                                                used in Fig 2 title + Fig 3
                                                error bar)
                         260.44 ± 18.13 km/s  (OLD reduction CLEAN, wide arc-masked,
                                                SECOND-reduction cross-check)
                         265.76 ± 17.87 km/s  (NEW reduction LEGACY un-cleaned, ref)
                         254.85 ± 17.87 km/s  (OLD reduction LEGACY un-cleaned, ref)
                         267.95 ± 30.10 km/s  (narrow Ca H+K+G, OLD cube X-check)
σ_e budget (WIDE,M12)= stat ±4.6 ⊕ Ishape ±2.27 ⊕ mask ±6.65
                       ⊕ centering ±4 ⊕ window ±3.82
                       ⊕ reduction ±3.45 ⊕ R_e-source ±6.13
                       (M11 cube-matched re-derivation + M12 D7 fold-in;
                        sys ±10.87, total ±11.77)

R_e                    = 2.305  = 16.23 kpc  (HST F140W+F200LP CoG mean;
                          FLAG A3c: raw CoG non-convergent, method sys ±0.08 )
log(M★/M⊙)              = 11.16 +0.32 / −0.08  (principled IR-extended masking, 10%;
                          supersedes 11.33 aperture)
                         11.04 ± 0.14         (20% flux err; fill-in reach 11.46)

z_deflector            = 0.67564                (NIST air rest λ, nb04)
```

```
z_source             = 1.302                    (AGEL DR2; cross-checked via
                                                 Fe II UV multiplet in KCRM
                                                 blue arm, I.2.2)
```

III.2 Methodology choices worth highlighting in the manuscript

1. Wide arc-masked window with explicit source-emission catalog. The single most consequential methodological choice. Halved the total σ_e uncertainty (±30 → ±18) and collapsed SPS-template spread by 6×. First documented catalog of source-emission contamination at a strong- lens deflector; reusable for any AGEL target with known z_source.
2. Frame-aware ppxf (per-SPS vacuum/air). Fixed a hidden bug that produced a −90 km/s FSPS V_sys offset throughout the nb07c era. The σ headline shifted by only +0.5 km/s, but the V_sys split-track collapsed 110 → 15 km/s and the §7 cross-check uncertainties halved.
3. Sersic2D bound-fix. Prevented the n=0.30 ellip=0.00 escape on the F200LP fit (a flat-disk pathology, not a physical solution). Removed a +20 km/s outlier in the I-shape sweep; the I-shape budget shrank from ±5.4 to ±1.5 km/s at the wide window.
4. Quadrature-conservative budget with literature-standard estimator. §6cum (cumulative I-weighted aperture ppxf) matches Cappellari et al. 2006 (SAURON IV §2.3) eq. 1 and Greene+20 ARA&A; three independent estimators (§7, §7b, arc-subtraction) agree within budget.
5. Aperture-vs-Sersic-total mass sensitivity. Quote the aperture M★ as the headline with the Sersic-total +16% as a stated floor — transparent about the aperture-photometry choice.

III.3 Caveats explicitly retained in the paper

1. No central σ. σ_e at R<R_e/8 was dropped (aperture inside half the seeing FWHM, only 3 spaxels). We report σ_e(<R_e/2) as a gradient diagnostic but do not quote a central value.
2. NIR SED residual. Bagpipes underpredicts F140W and F150W2 by ~8–11%; a more flexible SFH would likely tighten this. Effect on M★ already covered by quoted uncertainty.
3. KCWI cube header mislabel. Cite the multi-night combine with K409 Aug 30 UT 2025 as canonical; do *not* cite the header's "DATE-OBS = 2025-11-17, PROGNAME = U002, PROGPI = Jones" (last stacking pass only).
4. F140W vs F200LP centroid offset = 0.36″. Exceeds the 0.2″ sanity threshold; driven by the F200LP arc. HST-mean is robust because F140W is arc-clean.
5. One telluric residual masked alongside the source-emission bands. The def-rest 4525–4545 Å mask (obs 7593–7626 Å) is the leading edge of the O_2 atmospheric A-band, not source emission — confirmed 2026-05-18 (`NOTES_4534A_spike_investigation_2026-05-18.md`). The mask is unchanged; only the attribution was corrected.

III.4 What is *not* in this paper but is in the repo

- `notebooks/11_perbin_perspaxel_kinematics_wide.ipynb` — resolved σ map (per-spaxel + PowerBin) at R<1.5 R_e. Available as supplementary material if the referee asks for a kinematic map; the 17-spaxel S/N≥5 set shows the expected elliptical-bulge σ(R) gradient.
- `notebooks/04a_companion_redshifts.ipynb` — companion-galaxy redshifts in the same KCWI cube; outside scope but kept for the cluster-membership question.
- `notebooks/10_ETG_Mstar_radius_literature.ipynb` — comparison of AGEL0206 to local-ETG M★–R_e samples; supplementary context.

III.5 Open items before submission

(See `project_roadmap.md` for the current task list.)

0. Hδ targeted treatment (TODO 2026-05-26). The new `_clean` production pipeline removes the default ppxf ±800 km/s mask around the Balmer absorption lines (Hδ 4101.74, Hγ 4340.47, Hβ 4861.33) because they are

absorption-dominated, not emission, in this passive deflector. Hδ specifically may still need a targeted clip — its absorption-line shape is more template-sensitive than Hγ/Hβ (higher-order Balmer line, sensitive to template Balmer-decrement mismatch). To verify: re-run the local-MAD test (M5) on the no-Balmer-mask ppxf residuals and see whether Hδ shows up as a flagged region. Possible follow-ups: (a) keep all Balmer unmasked as in current production, (b) mask only Hδ at the ppxf-default ±800 km/s width, (c) mask Hδ at a narrower ±300 km/s width to keep most of its absorption signal while suppressing template mismatch. Flagged as inline comment in `scripts/bootstrap_ppxf.py: _determine_goodpixels_no_balmer`.

1. Manuscript kinematics section still needs rewriting around the wide arc-masked window headline.
2. K409 PI for the Aug 30 night needs to be confirmed (paper acknowledgements / observing-program citation).
3. Per-night DIMM seeing for the K409 Aug 30 night needs to be extracted from the observing log (currently using GUIDFWHM = 1.27″ from the Nov 17 stacking-pass night as a conservative estimate).
4. ~~Asymmetric posterior errors on Figure 3.~~ DONE 2026-05-19. Both Figures 2 and 3 now use the asymmetric pooled posterior: σ_e(<R_e) = 254.85 −18.20 / +17.59 km/s (total, stat from posterior + sys ±16.81 in quadrature each side). Stat alone: −7.0 / +5.2 km/s. Old symmetric ±17.87 lines kept commented in the figure cells (`fig2_wide`, `d55da320`) for traceability, with a dated update marker.
5. R_e/8 caveat phrasing. Add a one-sentence paper note that the aperture was dropped because it sits inside the seeing FWHM/2, referencing the seeing measurement (open item above).
6. Blue-arm Fe II source-z cross-check artifacts. The work described in I.2.2 was performed but the notebook/script and the measured z_source ± uncertainty are not yet committed in the repo. Once committed, replace the `[TBD]` placeholder in I.2.2 with the path + value.

Document index

- **`TESTS_AND_DIAGNOSTICS.md`** — the master test catalog (every individual test A1–L3 with status, paths, and cache locations).
- **`NOTES_nb09_frame_fix_and_final_sigma_e_2026-04-28.md`** — long-form derivation of the vacuum/air fix and the four methodology audits.
- **`HANDOFF_wide_arcmask_complete_2026-05-13.md`** — wide arc-masked window adoption + paper Figure 2 update.
- **`PROJECT_BRIEF.md`** — self-contained designer/agent handoff (still cites narrow-window 268 ± 32; pre-wide-window).
- **`CLAUDE.md`** — repository conventions and current headline summary.
- Memory index at `~/.claude/projects/-Users-rosador-Documents-AGEL-AGEL0206-kinematics-and-photometry/memory/MEMORY.md` for compressed cross-session references (project_, *reference_*, feedback_*).

Selected literature references

- Cappellari & Emsellem (2004) — ppxf
- Cappellari et al. (2006, SAURON IV) — §2.3 co-add-then-fit σ_e (cumulative I-weighted); NB its eq. 1 is the aperture correction, not σ_e. σ_e integral = Gültekin 2009 eq. 1
- Cappellari (2017, 2023) — ppxf revisions, high-z fwhm_gal_dict pattern
- Ciddor (1996) — vacuum-to-air conversion
- Kormendy & Ho (2013) — M•–σ scaling for ellipticals
- Greene et al. (2020 ARA&A) — σ_e aperture + M•–σ updates
- Gültekin et al. (2009) — discrete annular σ_e sum
- Carnall et al. (2018, 2019) — Bagpipes
- Bezanson et al. (2018); van der Wel et al. (2021) — LEGA-C, closest literature precedent for the wide-window rest 3500–5500 Å range
- AGEL DR2 (Carleton et al., paper in prep)